

Participation in research projects

As part of a research project under the guidance of Cand. Tech. Sci. Timofeev K. N., I designed, assembled, and programmed a portable infrared (IR) sensor for non-invasive skin diagnostics. The device functions as an external sensor for the software (SW) that forms the core of my bachelor's thesis titled "Optical Skin Diagnostic System Using Artificial Intelligence". Operating in the infrared spectrum (wavelength: 880 nm), the sensor is equipped with a NodeMCU (ESP8266) microcontroller and transmits data to a PC/smartphone via WiFi interface.

My responsibilities included:

- Designing the sensor circuit and optical system,
- Calibrating IR emitters and photodetectors,
- Developing firmware for device control and signal preprocessing,
- API integration with the software.

A patent application was prepared based on the device; however, due to time constraints for the thesis project and prioritization of AI algorithm development, patenting the hardware component was postponed.